CURRICULUM VITAE

Name: Bershtein, Olga.

Date of birth: September 29, 1981.

Citizenship: Ukraine.

Education:
Undergraduate: Bachelor in Mathematics, Kharkov National University, 2002.
Graduate: Magister in Mathematics, Kharkov National University, 2003.
Postgraduate: PhD student of Institute for Low Temperature Physics and Engineering, 2003-2006.
Affiliation:
Junior research fellow, Institute for Low Temperature Physics and Engineering, from November 2006 till December 2008.
Research fellow, Institute for Low temperature Physics and Engineering, from January 2009.

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Fields of research: Quantum groups and quantum homogeneous spaces, representation theory of real reductive groups and their quantum analogues, noncommutative harmonic analysis and special functions.

I investigate quantum groups and quantum bounded symmetric domains together with noncommutative analogs of algebras of functions related to them. Also, I consider quantum Harish-Chandra modules, principal series of representations realized in these domains together

with noncommutative harmonic and complex analysis appeared in the domains.

Fields of interests: Quantum groups and quantum homogeneous spaces, representation theory, noncommutative harmonic analysis, Kac-Moody algebras.

Awards:

Scholarship of N.I. Akhiezer's fund (2008).

Scholarship of the fund "Kharkov Hope" , N.I. Akhiezer's award (2008).

Scholarship of the National Academy of Sciences of Ukraine for young scientists (October 2008-October 2010).

Publications:

1. O.Bershtein *Degenerate principal series of quantum Harish-Chandra modules* – J. Math. Phys. **45** vol. 10, (2004), 3800-3827; math.QA/0406343.

2. O.Bershtein *Regular functions on the Shilov boundary* – J. Algebra and Applications, vol.4, **6** (2005), 613-629; math.QA/0411118.

3. O.Bershtein On some bounded symmetric domains – Proceedings of Lie Theory and Applications-6, (2005), 150-156; math.QA/0512262.

4. O.Bershtein, Ye.Kolisnyk, L.Vaksman *On a q-analog of the Wallach-Okounkov formula* – Letters in Math.Phys., **78**, No. 1 (2006), 97-109; math.QA/0603081.

5. O.Bershtein *On a q-analog of a Sahi result* – J. of Math. Phys., **48**, vol. 4, (2007), 1917-1924; math.QA/0606215.

6. O.Bershtein, A.Stolin, L.Vaksman *Spherical principal series of quantum Harish-Chandra modules*, – J. of Math. Phys., Anal. and Geom., **3** No. 2, (2007), 157-175; math.QA/0609687.

7. O.Bershtein, S.Sinelshchikov, A q-analog of the Hua equations, – J. of Math. Phys., Anal. and Geom. (accepted for publishing) (2009).

8. O.Bershtein, Ye.Kolisnyk, Plancherel formula for the quantum matrix ball-I, – J. of Math. Phys., Anal. and Geom. (accepted for publishing) (2009).

Conferences:

1. 14-16 June 2004, *First Karazin's Nature Studies*, Kharkiv, Ukraine (session talk).

2. 22-27 June 2004, Satellite Conference to 4th European Mathematician Congress *Representation Theory and its Applications*, Uppsala, Sweden (session talk).

3. 15-21 August 2005, *Quantum Theory and Symmetries*, Varna, Bulgaria (session talk).

4. 24-30 June 2007, *Symmetry in Nonlinear Mathematical Physics*, Kiev, Ukraine (session talk).

5. 6-8 August 2007, Workshop on *Quantum Groups and Noncommutative Geometry*, Bonn, Germany (poster presentation).

6. 17-22 December 2007, *Transformation Groups*, Moscow, Russia (poster presentation).

7. 21-27 July 2008, Satellite Conference to 5th European Mathematician Congress *Noncommutative Structures in Mathematics and Physics*, Brussels, Belgium (plenary talk).